Pituitary Disorders I: Hypothalamus–Pituitary, Tumours, Prolactinoma & Acromegaly

Internal Medicine Department | Source: `9- hypothalamus and pituitary – galactorrhea, acromegaly` (57 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.
>
> 🔗 **Bridge from earlier lectures:**
>
> - **Physiology (regulatory systems, hormones):** the hypothalamic–pituitary axis is the textbook **long and short negative-feedback loop**.
> - **Thyroid 2: primary hypothyroidism → ↑ TRH → ↑ prolactin** (this lecture's first case).
> - **Hyperparathyroidism: MEN 1** = **P**ituitary + **P**arathyroid + **P**ancreas.
> - **Hypertension:** acromegaly was a cause of secondary HTN.
> - **Lung cancer: bronchial carcinoid** can secrete **GHRH** (ectopic acromegaly).

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology and etiology of pituitary disorders	Hormone ↔ cell type; feedback
Knowledge	⭐ Epidemiology, manifestations, complications, management	Tumour types %, mass effects, prolactinoma/acromegaly features
Skill	⭐ Interpret manifestations + investigations → diagnosis	IGF-1 → OGTT; prolactin < 200 = not prolactinoma
Skill	Evidence-based management	Dopamine agonist first for prolactinoma; surgery first for GH adenoma
Skill	Recognise emergencies	Apoplexy; visual loss

Lecture flow: Pituitary anatomy (site, weight, blood supply) → Histology → Regulation → **Pituitary tumours** (incidence, endocrine and mass effects) → **Hyperprolactinaemia** (physiology, causes, prolactinoma: features, investigations, treatment) → Initial treatment of adenomas → **GH hypersecretion** (physiology, gigantism, DDx of tall stature, **acromegaly**: features, investigations, treatment) → Cases.

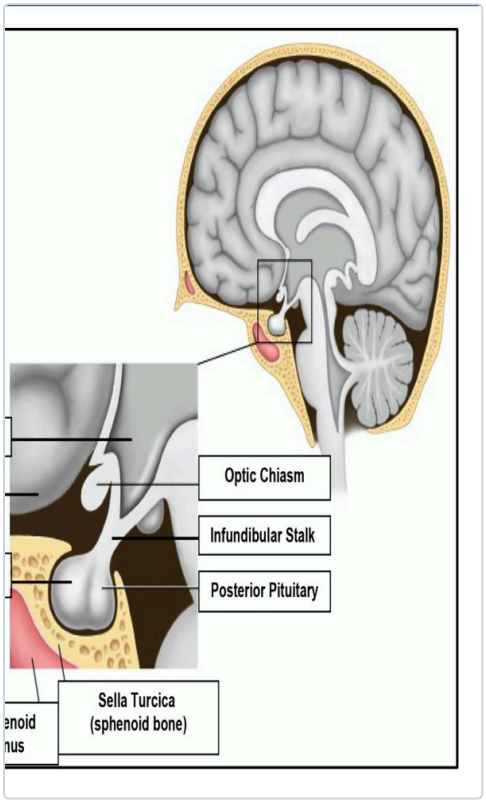
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2. Main Content (Lecture Order)

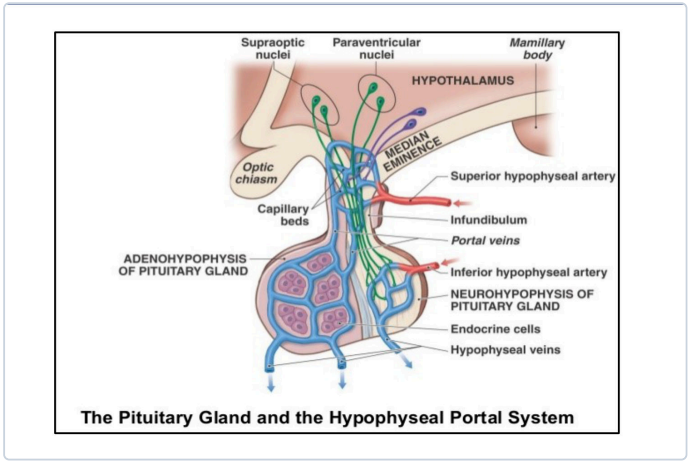
2.1 Anatomy (Slides 4–8)

Feature	Detail
Site	In the **sella turcica**. Dura lines the sella and forms a roof with a hole (**diaphragma sellae**) through which the **pituitary stalk** connects to the hypothalamus
Weight	**½–1 g** (6–13 mm in each dimension)

Feature	Detail
Hypothalamus blood supply	**Superior hypophyseal artery** (branch of the **ICA**)
⭐ **Anterior pituitary**	**Hypophyseal portal circulation**: hypothalamus → **long portal veins** (the **main blood supply** of the anterior lobe) → capillary plexuses → **short portal veins** (main venous drainage)
Posterior lobe	**Inferior hypophyseal artery**



Slide 6: sella turcica and relations



Slide 8: hypophyseal portal system

[added] Because the anterior lobe depends on a low-pressure portal (venous) supply, it is vulnerable to infarction: Sheehan syndrome (next lecture).

2.2 Histology (Slides 9–11) ⭐

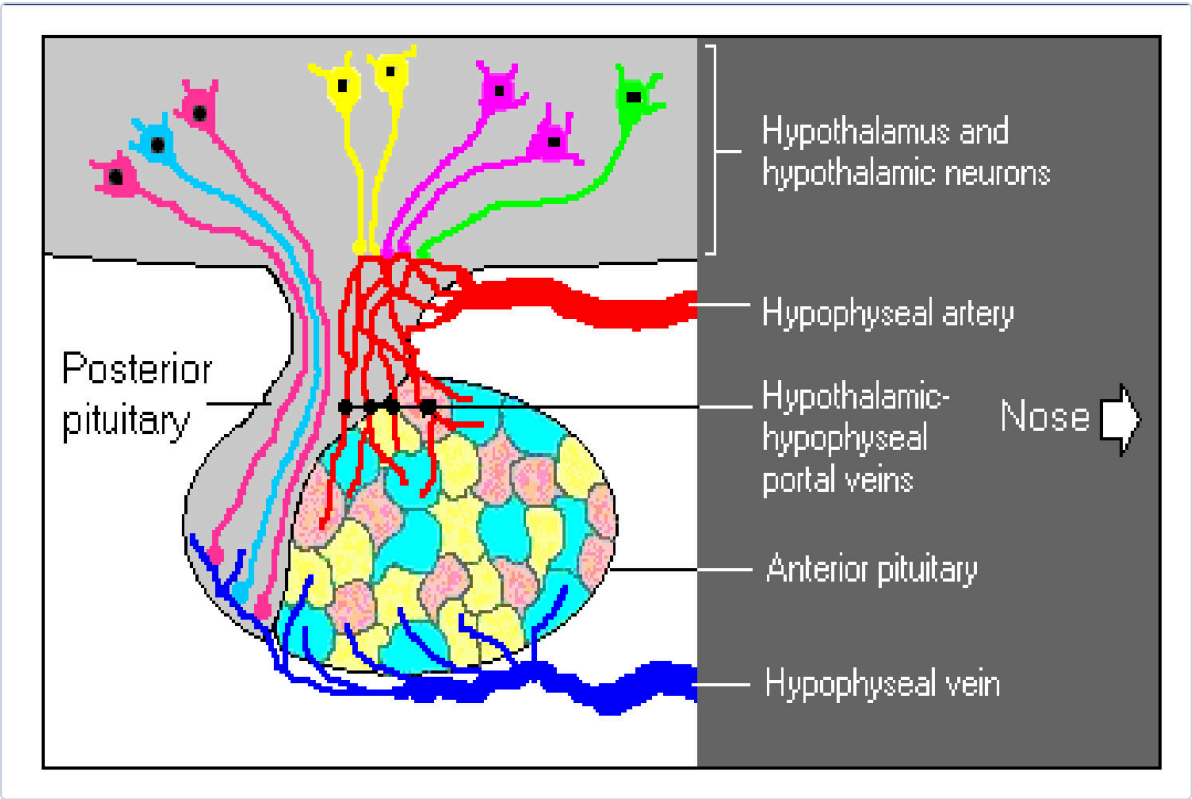
Cell type (stain)	Hormones
Acidophils	**Prolactin**, **GH**
Basophils	**TSH, ACTH, FSH** (→ follicle maturation and spermatogenesis), **LH** (→ ovulation, corpus luteum, ↑ testosterone)
Chromophobes	**No secretion**

> **Mnemonic: "GPA"** → **G**H and **P**RL are **A**cidophils. **B-FLAT** → **B**asophils make **F**SH, **L**H, **A**CTH, **T**SH.

Posterior pituitary hormones are made in the hypothalamus, then transported down the axons to the posterior lobe:

- **ADH**: **supraoptic** nucleus
- **oxytocin**: **paraventricular** nucleus

(The slide's "respectively" is in the wrong order. See §8.)



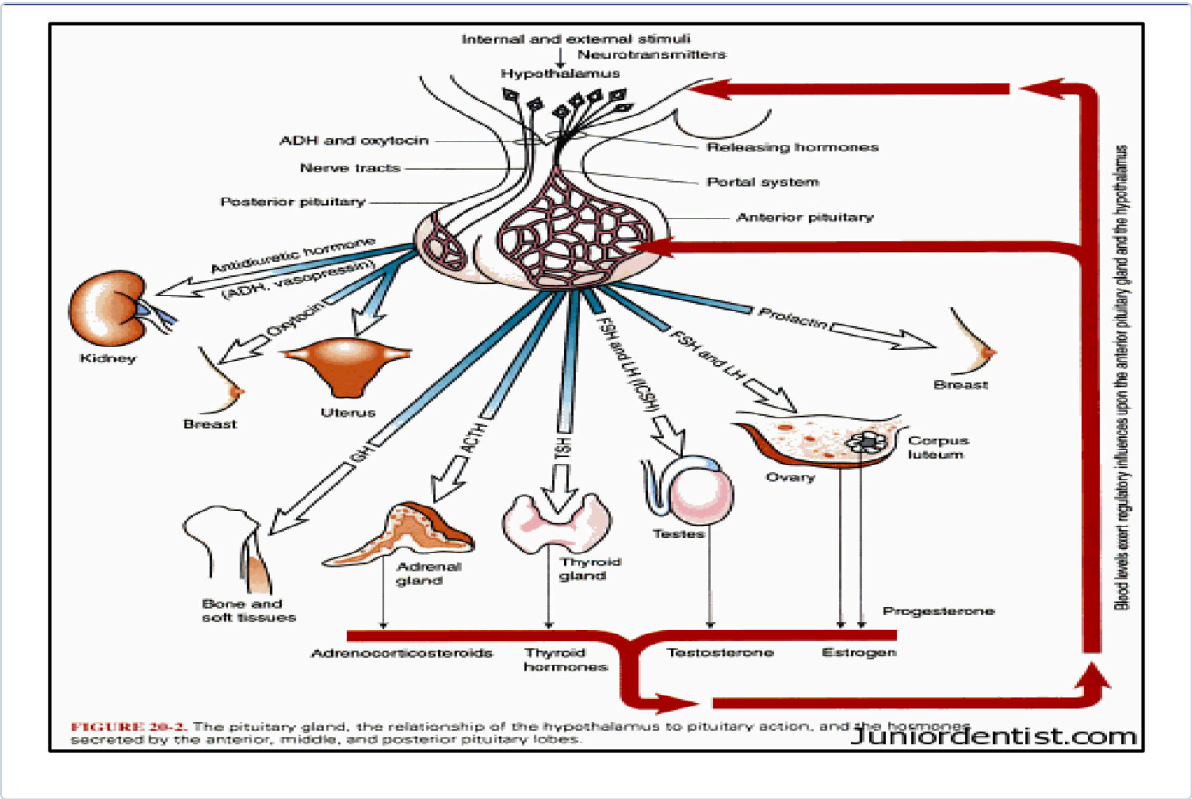
Slide 11: hypothalamic–hypophyseal portal veins

2.3 Regulation of Anterior Pituitary Secretion (Slides 12–13) ⭐

Hypothalamic hormone	Effect
TRH	↑ **TSH** and ↑ **PRL**
GnRH (LHRH)	↑ **LH, FSH** (**LH > FSH**)
GHRH	↑ **GH**
⭐ **GHIH (somatostatin)**	↓ **GH**. Also in pancreatic **D cells**: ↓ insulin, glucagon, gastrin, VIP
CRH	↑ **ACTH**
⭐ **Dopamine**	**Prolactin has no specific releasing factor**. **Dopamine is the PRL-inhibiting factor** (tonic inhibition)

Feedback:

- **Long loop:** target-gland hormones act on the pituitary and hypothalamus (e.g., **thyroxine blunts the TSH response to TRH**).
- **Short loop:** the pituitary hormone acts on the hypothalamus.



Slide 13: hypothalamic–pituitary–target organ map

> 🔁 **Recap: Anatomy and physiology.**
>
> - **Gland:** in the sella, below the diaphragma sellae, ½–1 g. Anterior lobe fed by **portal veins**; posterior by the inferior hypophyseal artery.
> - **Cells:** acidophils = GH, PRL; basophils = TSH, ACTH, FSH, LH.
> - **Hypothalamic control:** TRH (TSH + PRL), GnRH, GHRH, somatostatin (↓GH), CRH. PRL is under **dopamine inhibition**. Long and short feedback loops.
>
> 🔗 **Link:** An adenoma in this small box causes problems in two ways: the hormone it makes, and what it squeezes.
>
> ▶️ **Up next: Pituitary tumours.**

2.4 Pituitary Tumours (Slides 14–19) ⭐

Incidence: about **15% of intracranial tumours**.

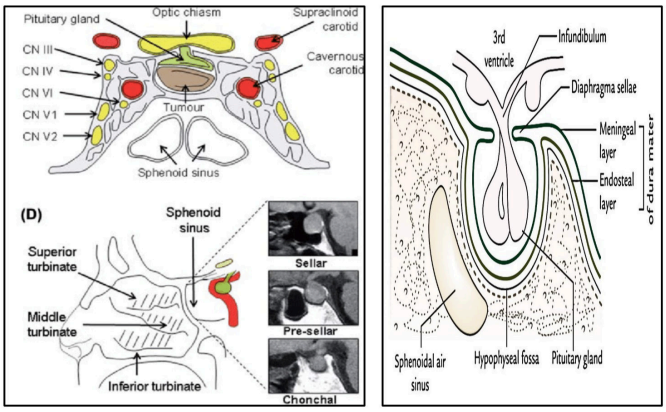
Type	%
⭐ **Prolactin-producing**	**35%** (commonest functioning)
Non-functioning	**30%**
GH-secreting	**20%**
Mixed PRL + GH	**10%**
TSH, LH, FSH, ACTH	**5%**

Clinical picture:

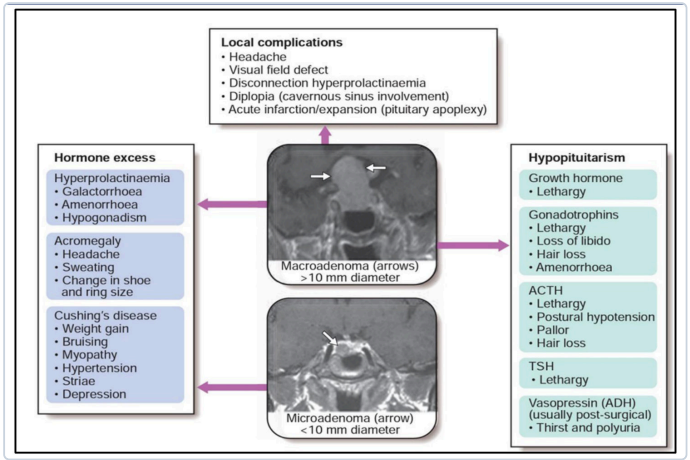
- **Acute:** ⭐ **pituitary apoplexy**.
- **Chronic, endocrine:** - **hyperpituitarism** (by the hormone secreted) - **hypopituitarism** - association with **MEN 1** (pituitary, pancreas, parathyroid)

- **Chronic, mass effects:**

Direction	Effect
Headache	⭐ **Bitemporal**. It **improves** when the tumour erodes the clinoid process and escapes into the cranial cavity. Rare and late: **occipital headache + ↑ ICP**
⭐ **Superior**	**Chiasmal syndrome**: ↓ visual acuity + ⭐ **bitemporal hemianopia**. **Hypothalamic syndrome**: **DI**, disturbed thirst, temperature, appetite and sleep; **SIADH**
Lateral (cavernous sinus)	**CN III, IV, VI and V1 → diplopia + facial pain**; temporal lobe dysfunction
Inferior	**Nasopharyngeal mass**, ⭐ **CSF rhinorrhoea**
Posterior	**Bilateral pyramidal (Δ) signs**



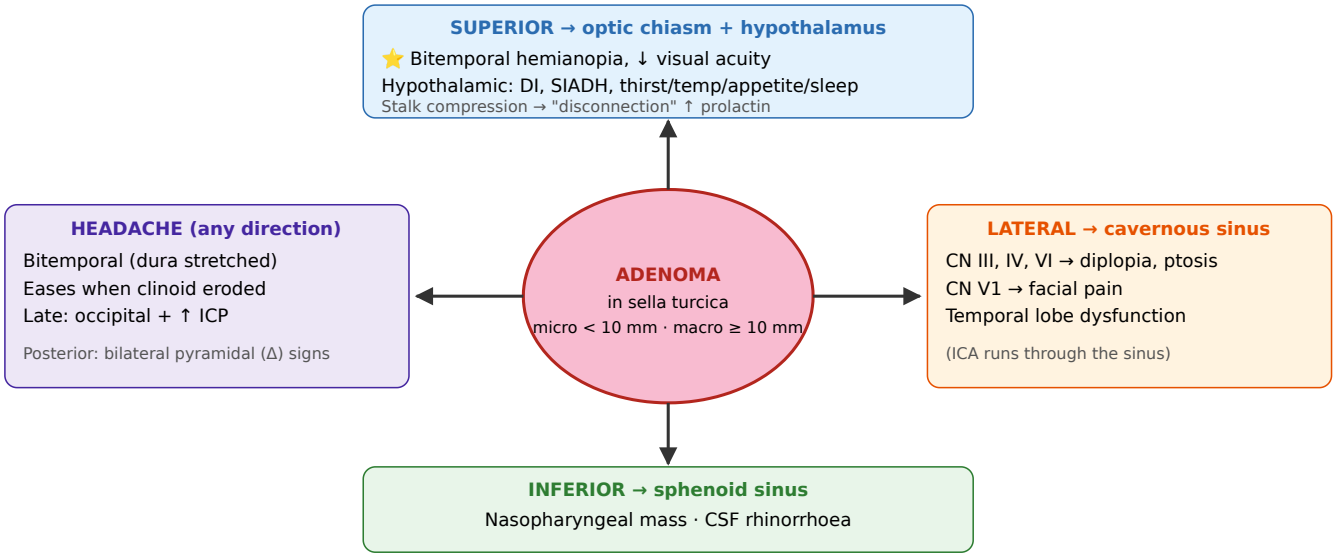
Slide 18: cavernous sinus nerves and sellar anatomy



Slide 19: local effects, hormone excess and hypopituitarism

Slide 19 summary:

- **Local:** headache, visual field defect, **disconnection hyperprolactinaemia**, diplopia, apoplexy.
- **Excess:** PRL (galactorrhoea, amenorrhoea, hypogonadism); acromegaly (headache, sweating, change in shoe and ring size); Cushing's (weight gain, bruising, myopathy, HTN, striae, depression).
- **Hypopituitarism:** - GH: lethargy - gonadotrophins: lethargy, loss of libido, hair loss, amenorrhoea - ACTH: lethargy, postural hypotension, pallor, hair loss - TSH: lethargy - ADH: usually post-surgical; thirst and polyuria
- **Size: macroadenoma > 10 mm**; **microadenoma < 10 mm**.



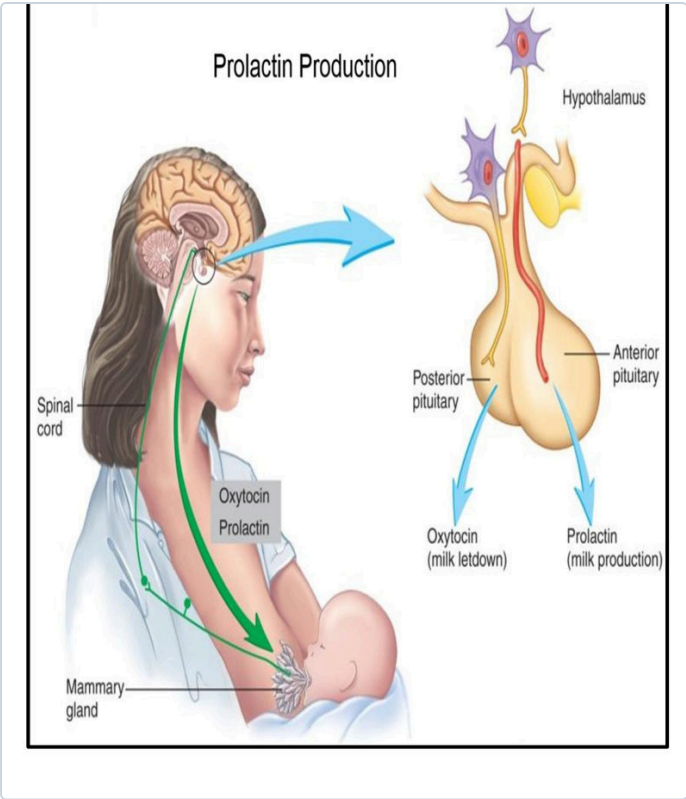
Added diagram: direction of growth and mass effects

> 🔁 **Recap: Tumours.** 15% of intracranial tumours. PRL 35% > non-functioning 30% > GH 20% > mixed 10% > others 5%. Endocrine excess or deficiency; MEN 1. Mass effects: bitemporal headache; up → bitemporal hemianopia, DI/SIADH; lateral → CN III/IV/V1/VI; down → CSF rhinorrhoea; back → pyramidal signs. Acute bleed = apoplexy.
>
> 🔗 **Link:** The commonest functioning tumour secretes prolactin, but most high prolactin levels are **not** tumours.
>
> ▶️ **Up next: Hyperprolactinaemia.**

2.5 Hyperprolactinaemia (Slides 20–21) ⭐

Physiology:

- **Actions: breast milk production**; **suppression of menstruation**.
- **Control: nocturnal peak** (sleep-related).



Slide 20: prolactin and oxytocin production

Causes (slide 21):

Group	Causes
A. Physiological	⭐ **Pregnancy** · **"3 S"**: **S**uckling, **S**tress, **S**timulation of the nipple
B. Drugs	**Dopamine-depleting** (methyldopa, reserpine) · ⭐ **dopamine-receptor antagonists** (**phenothiazines/chlorpromazine**, **antipsychotics**; *[added]* metoclopramide, domperidone, risperidone) · **oestrogens** · **opiates**
C. Pathological: "3C, 3P"	**C**RF (chronic renal failure) · **C**irrhosis · **C**hest wall trauma (*[added]* zoster, burns) · **P**ituitary (prolactinoma; **tumours compressing the stalk**) · **P**roduction by other tumours (paraneoplastic) · ⭐ **P**rimary **hypothyroidism** (↑ TRH)

2.6 Prolactinoma (Slides 22–27) ⭐

Size: usually a **microadenoma in women** and a **macroadenoma in men** *[added: women present early with menstrual change; men present late with mass effects]*.

Group	Manifestations (↑ PRL → ↓↓ GnRH)
Sellar tumour	Mass effects (above)
Females	⭐ **Amenorrhoea and/or galactorrhoea** · **hypo-oestrogenic**: ↓ libido, ↓ vaginal secretion, dyspareunia, **osteoporosis** · **hyperandrogenic**: hirsutism, acne (↑ PRL → ↑ adrenal androgens)
Males	⭐ **↓ libido and impotence** · oligospermia, infertility · **gynaecomastia (rare)**
Both	**Delayed puberty**

Investigations (slide 24):

1. **Basal prolactin:** normal ♀ **0–25 ng/mL**, ♂ **0–20 ng/mL**. - ⭐ **Clinical pearl: PRL < 200 ng/mL suggests a cause other than prolactinoma** (drugs, primary hypothyroidism). - ⭐ Always **review the medication history**.

2. **X-ray, CT, MRI** to detect pituitary tumours. - ⭐ **Clinical pearl:** order **hormone assays BEFORE imaging** to avoid overdiagnosing pituitary **incidentalomas**.

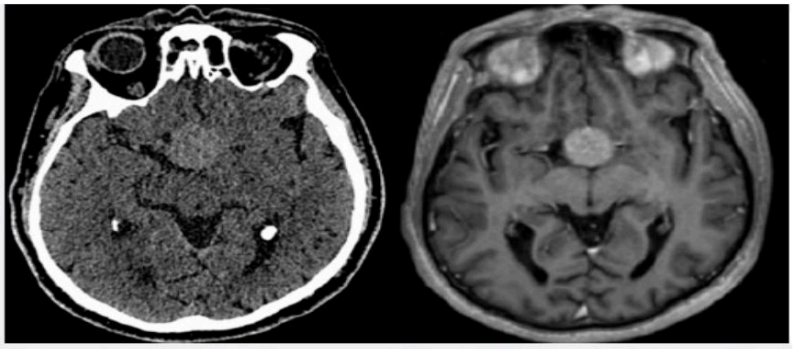
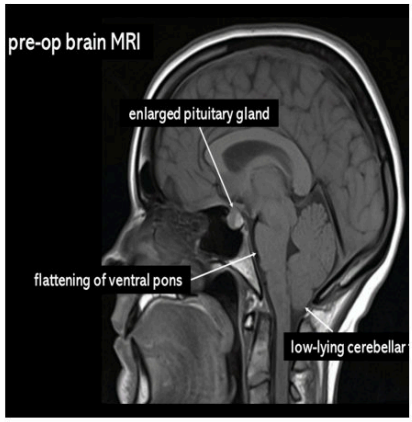


Figure 1: Treatment planning CT and MR images of a patient with pituitary adenoma.



Slide 24: CT/MR of pituitary adenoma

[added] Also: pregnancy test, TSH, renal and liver function. Beware the "hook effect": a huge macroprolactinoma can give a falsely modest PRL, so dilute the sample. Macroprolactin is biologically inactive.

Treatment (slides 25–27):

Option	Detail
1. Surgery	**Trans-sphenoidal** for **microadenoma (< 10 mm)**. Complications: **hypopituitarism, CSF rhinorrhoea, meningitis, optic nerve damage**. **Trans-frontal** for large tumours (> 10 mm)
2. Radiotherapy	**4500 rads**
⭐ **3. Medical: dopamine agonist**	**Bromocriptine**. Aims: ↓ hyperprolactinaemia **from any cause**; **restores menses and fertility**, stops galactorrhoea; ⭐ **shrinks the tumour**. Dose **2.5 mg tds with meals** for 1–2 years. Side effects: **nausea, vomiting, postural hypotension**. ⭐ **Cabergoline** is more effective (but expensive)

Initial treatment of pituitary adenomas (slide 27):

Tumour	First line
⭐ **Prolactinoma** (symptomatic or macroadenoma)	**Pharmacotherapy** (dopamine agonist)
Secretory adenomas other than prolactinoma (GH, ACTH, TSH)	⭐ **Surgery**
Symptomatic non-secretory adenoma	**Surgery**
Asymptomatic microprolactinoma	**Observation**
Asymptomatic non-secretory adenoma	**Observation**

Tumor type	First-line treatment
Prolactinomas (symptomatic or macroadenomas)	Pharmacotherapy
Secretory adenomas (except prolactinomas)	Surgery
Symptomatic nonsecretory adenomas	
Asymptomatic microprolactinomas	Observation
Asymptomatic nonsecretory adenomas	

Slide 27: initial treatment of pituitary adenomas

🔁 **Recap: Prolactin.** Dopamine inhibits PRL; TRH stimulates it.

- **Causes:** pregnancy, the 3 S's; drugs (antipsychotics, methyldopa, oestrogen, opiates); 3C (CRF, cirrhosis, chest wall), 3P (pituitary/stalk, paraneoplastic, primary hypothyroidism).
- **Prolactinoma:** ♀ micro (amenorrhoea, galactorrhoea, osteoporosis, hirsutism), ♂ macro (impotence, mass effects).
- **Tests:** PRL < 200 → look for another cause; hormones before imaging.
- **Treatment:** dopamine agonist first (cabergoline > bromocriptine; shrinks the tumour). Surgery or radiotherapy if that fails.

🔗 **Link:** The other acidophil hormone, GH, produces the most visible endocrine disease.

▶️ **Up next: GH excess (gigantism and acromegaly).**

2.7 GH Physiology (Slides 28–30)

- **Actions:** ↑ **blood glucose**, ↑ **protein synthesis**, ↑ **free fatty acids**.
- Acts via **somatomedins (IGF-1)**, made mainly in the **liver**.

↑ GH (via GHRH)	↓ GH (via somatostatin)
Hypoglycaemia · stress, exercise · oestrogen, **clonidine, L-dopa**	⭐ **Hyperglycaemia** · obesity

GH excess comes from an **acidophil adenoma**:

- **Gigantism**: before **epiphyseal fusion**.
- **Acromegaly**: after epiphyseal fusion.

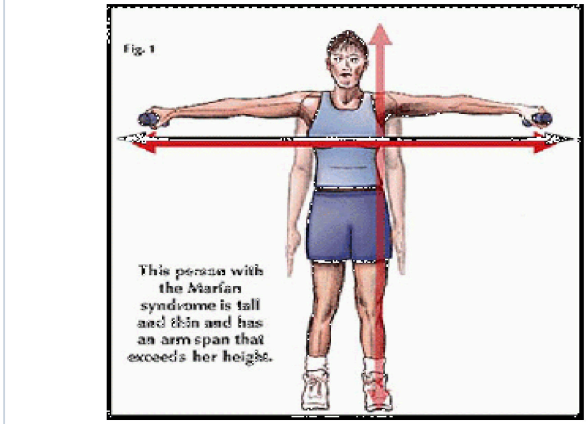
2.8 Gigantism & Tall Stature (Slides 31–35)

- **Massive skeletal growth** (up to **2.5 m**); soft tissues also enlarge.

- **Early:** strong. **Later ("decline stage"):** weakness, **hypogonadism, myopathy, peripheral neuropathy**.



Slide 32: pituitary gigantism



Slide 35: Marfan arm span > height

DDx of tall stature (slide 34):

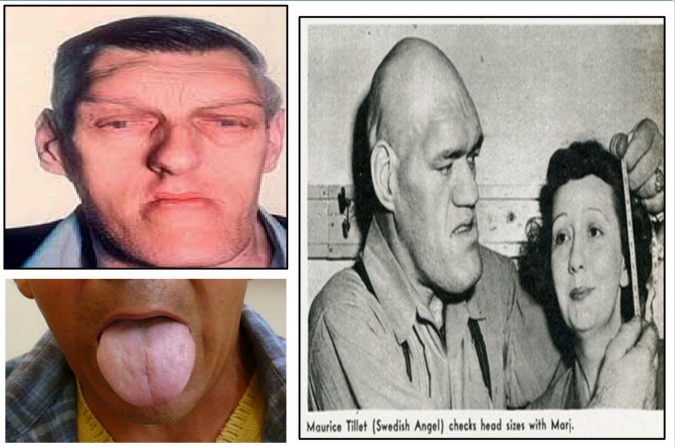
Cause	Clue
⭐ **Constitutional / genetic (most common)**	**Proportionate**, **normal GH**
Cerebral gigantism (Sotos)	**Early puberty**
Eunuchoidism	**Disproportionate** (long limbs) + hypogonadism
Hyperthyroidism	
Homocystinuria	Marfan-like + **mental retardation** (*[added]* downward lens dislocation, thrombosis)
Marfan	Arm span > height, upward lens dislocation *[added]*
⭐ **Klinefelter**	**Azoospermia, gynaecomastia, positive buccal smear** (Barr body; 47 XXY)

2.9 Acromegaly: Clinical Picture (Slides 36–44) ⭐

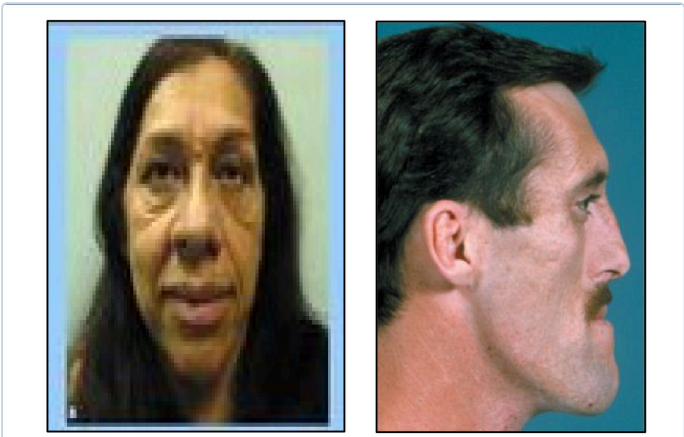
- **Acral parts** (hands, nose, feet) enlarge.
- **Slowly progressive**, starting in the **2nd–4th decades**. **Manifestations appear 10–20 years later** (late diagnosis).

#	System	Features
1	**Face**	Enlarged lips, nose and soft tissue; ⭐ **prognathism** with ↑ spacing of lower teeth; ⭐ **macroglossia**; **frontal bossing** (big frontal sinuses); **cavernous, husky voice**
2	**Soft tissue**	⭐ **↑ ring, glove, shoe and hat size**; **spade-like hands**
3	**Neuro**	⭐ **Carpal tunnel syndrome**; peripheral neuropathy; diabetic neuropathy; **proximal myopathy**; tumour mass effects

#	System	Features
4	**Visceromegaly**	Liver, spleen, kidney…
5	**Joints**	**Osteoarthritis** (spine, hip, knee), late
6	**CVS**	⭐ **Hypertension**, **cardiomegaly**, Raynaud's, conduction defects and arrhythmia, ⭐ **sleep apnoea**
7	**Skin**	**Hypertrichosis**, ↑ pigmentation, **acanthosis nigricans**, **warm, moist, thick, oily skin** (acne)
8	**Metabolic**	↑ BMR → heat intolerance, **hyperhidrosis**; ⭐ **glucose intolerance 50%, frank DM 25%**; **hyperphosphataemia**; **hypercalcaemia 16%** (↑ $1,25(OH)_2D$)
9	**Endocrine**	Co-secretion: **PRL** (hyperprolactinaemia), **ACTH** (Cushing's), **TSH** (secondary thyrotoxicosis); **MEN 1**; **DI** (suprasellar extension)



Slide 39: acromegalic face, macroglossia



Slide 40: progression of coarse facial features



Slide 41: spade hands, big feet, teeth spacing

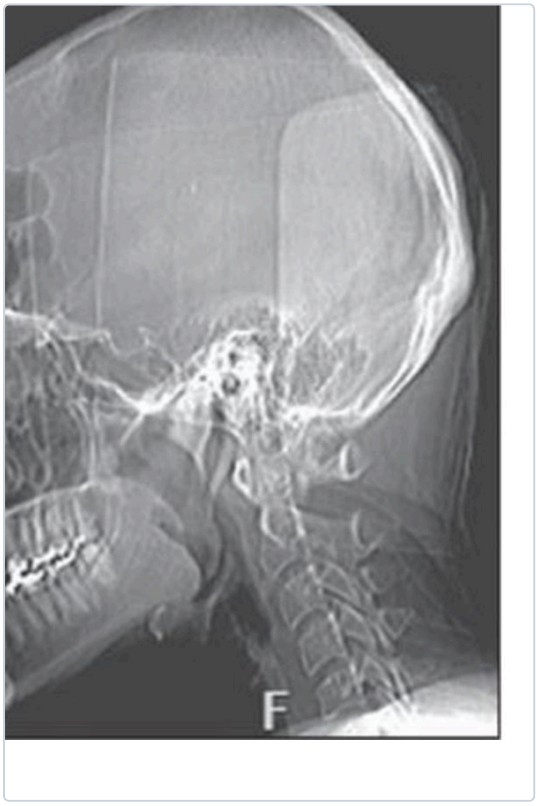
Mnemonic for acromegaly: "ABCDEF" *[added]*:

- **A**rthralgia / **A**rthritis
- **B**P ↑
- **C**arpal tunnel
- **D**iabetes
- **E**nlarged organs
- **F**ield defect (bitemporal)

2.10 Acromegaly: Investigations (Slides 45–48) ⭐

Radiology:

X-ray	Finding
Skull	Thick cortex; **pneumatised frontal sinus**; prominent occipital protuberance; **prognathism**, big mastoid; **sella: double floor (early) → ballooning + clinoid erosion**
Hands	⭐ **Tufting of terminal phalanges** ("brush border"/**mushroom**, from periosteal thickening) → pseudoclubbing
Heel	⭐ **Heel pad > 22 mm**
Joints	Osteoarthritis
CT/MRI	Pituitary tumour
CXR	Exclude **bronchial carcinoid** (ectopic GHRH)



Slide 45: acromegalic skull



Slide 46: tufted terminal phalanx

Eyes:

- Visual fields and ophthalmology: **bitemporal hemianopia**, papilloedema, optic atrophy, glaucoma.

- ⭐ **Caution:** visual defects can occur **even without visual symptoms** in patients with an adenoma, so test all of them.

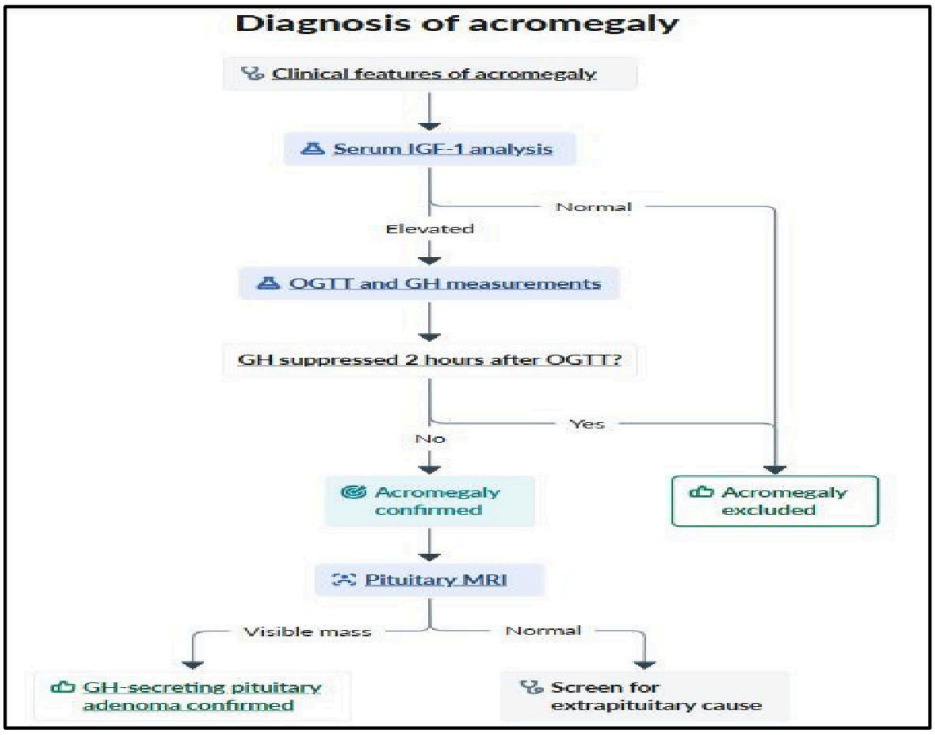
Endocrine:

Test	Role
Basal GH	↑ (> 6 ng/mL). ⭐ **Hard to interpret: GH is pulsatile**. *Random GH is not recommended for diagnosis*
⭐ **IGF-1**	↑ → **BEST INITIAL test**
⭐ **OGTT: GH fails to suppress (< 2 ng/mL)**	**MOST ACCURATE (confirmatory) test**
GHRH	↑ in **ectopic secretion (bronchial carcinoid)**

- **Metabolic tests:** ↑ BMR, hyperglycaemia, hypercalcaemia.
- ⭐ **Echocardiography + colonoscopy**: cardiomegaly and **colon polyps** (*[added]* ↑ colorectal cancer risk).

Diagnostic algorithm (slide 48):

1. Clinical features → **IGF-1**. - Normal → acromegaly **excluded**.
2. Elevated → **OGTT + GH**. Is GH suppressed at 2 h? - Yes → **excluded**. - No → **acromegaly confirmed**.
3. **Pituitary MRI.** - Visible mass → **GH-secreting adenoma**. - Normal → **screen for an extrapituitary cause**.



Slide 48: diagnosis of acromegaly

2.11 Acromegaly: Treatment (Slides 49–50)

Line	Detail
⭐ **1. Surgery**	**As for prolactinoma** (trans-sphenoidal), **first line** for GH adenoma

Line	Detail
2. Radiotherapy	Proton beam, heavy particles, supervoltage. **Takes several years** for the full effect
3a. Dopamine agonist (bromocriptine)	When surgery is contraindicated or rapid relief is needed. **15–30 mg/day** (higher than for prolactinoma). GH ↓ in **only 10–40%**. Used as **adjunct** after surgery or radiotherapy
3b. Somatostatin	Not specific for GH; needs an **infusion**
⭐ **3c. Octreotide** (somatostatin analogue)	**Longer acting, specific**, **SC every 12 h** (*[added]* monthly depot forms; lanreotide)

⭐ **Clinical pearl (slide 50): somatostatin analogues and dopamine agonists can reduce GH secretion and tumour size**. *[added]* **Pegvisomant** (a GH-receptor antagonist) normalises IGF-1 but **does not shrink the tumour**.

> 🔁 **Recap: Acromegaly.**
> - **Physiology:** GH acts via IGF-1 (liver). Hypoglycaemia stimulates GH; hyperglycaemia suppresses it.
> - **Gigantism vs acromegaly:** gigantism before epiphyseal fusion, acromegaly after. DDx of tall stature: constitutional is commonest; Klinefelter, Marfan, homocystinuria, Sotos.
> - **Acromegaly features:** coarse face, prognathism, macroglossia, spade hands, ↑ ring/shoe size, carpal tunnel, OSA, HTN, cardiomegaly, DM 25%, hyperhidrosis, oily skin, colon polyps.
> - **Diagnosis:** IGF-1 first → OGTT (no GH suppression < 2) confirms → MRI. Heel pad > 22 mm; tufted phalanges.
> - **Treatment:** trans-sphenoidal surgery → octreotide / dopamine agonist / radiotherapy.

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3. High-Yield Comparison Tables

Table A: Prolactinoma vs Acromegaly

	Prolactinoma	Acromegaly
Frequency	**35%** (commonest)	**20%**
Typical size	♀ micro, ♂ macro	Usually macro *[added]*
Key features	Amenorrhoea, galactorrhoea, infertility, ↓ libido	Coarse features, spade hands, OSA, DM, HTN
Best initial test	**Prolactin** (+ TSH, β-hCG, drugs)	**IGF-1**
Confirmatory	MRI (if PRL raised and secondary causes excluded)	**OGTT: GH not suppressed < 2 ng/mL**
First-line Rx	⭐ **Dopamine agonist** (cabergoline/bromocriptine 2.5 mg tds)	⭐ **Trans-sphenoidal surgery**
Second line	Surgery, radiotherapy	Octreotide, DA (15–30 mg/day), radiotherapy

Table B: Hyperprolactinaemia: Clue → Cause

Clue	Cause
Pregnant / breastfeeding	Physiological

Clue	Cause
On antipsychotic / metoclopramide / methyldopa	**Drug**: stop or switch
↑ TSH	**Primary hypothyroidism** (TRH) → levothyroxine
↑ creatinine	CRF (↓ clearance)
PRL **> 200** + sellar mass	**Prolactinoma**
PRL modestly ↑ + **large non-functioning** mass	**Stalk compression** (disconnection)
Chest wall scar / zoster	Neural reflex

Table C: Mass Effects by Direction

Direction	Structure	Sign
Up	Optic chiasm	**Bitemporal hemianopia**
Up	Hypothalamus	DI, SIADH, thirst/temp/appetite/sleep
Lateral	Cavernous sinus (III, IV, V1, VI)	Diplopia, ptosis, facial pain
Down	Sphenoid	**CSF rhinorrhoea**, nasopharyngeal mass
Back	Brainstem	Bilateral pyramidal signs
Any	Dura	**Bitemporal headache**

Table D: Tall Stature DDx

Cause	Proportion	GH	Distinguishing clue
Constitutional	Proportionate	Normal	Tall parents
Gigantism	Proportionate	**↑**	Soft-tissue enlargement, decline stage
Klinefelter	Long legs	N	Small testes, gynaecomastia, **XXY**
Marfan	Arm span > height	N	Lens up, aortic root, MVP
Homocystinuria	Marfanoid	N	**Mental retardation**, lens down, thrombosis
Eunuchoidism	Disproportionate	N	Hypogonadism
Sotos		N	**Early puberty**, macrocephaly
Hyperthyroidism		N	Thyrotoxic features

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Hypothalamic hormones **stimulate** the pituitary	**Prolactin is under tonic INHIBITION by dopamine**; stalk damage ↑ PRL
TRH ↑ TSH	It also ↑ **PRL**, which is why **primary hypothyroidism causes hyperprolactinaemia**
High PRL = prolactinoma	**PRL < 200** → think drugs, hypothyroidism, stalk compression
Pituitary tumours → surgery	⭐ **Prolactinoma → dopamine agonist FIRST**
Dopamine agonists for prolactin	Also used in **acromegaly** (higher dose; only 10–40% respond)
Random GH diagnoses acromegaly	**Pulsatile**: use **IGF-1** then **OGTT suppression**
Glucose suppresses GH	Failure to suppress = acromegaly

Rule	Exception / nuance
Headache from pituitary tumours worsens with growth	It may **improve** once the tumour erodes the clinoid and escapes the sella
Imaging first for a pituitary problem	**Hormones first**: incidentalomas are common
Galactorrhoea is the PRL sign	**Men** present with **impotence and mass effects**; galactorrhoea is rare
Tall = gigantism	**Constitutional tall stature is most common**

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5. Clinical Correlations

- **Woman with amenorrhoea + galactorrhoea, PRL 120, on risperidone/metoclopramide** → drug-induced; stop or switch the drug and recheck.
- **Woman with galactorrhoea, TSH 17, PRL 85** → **primary hypothyroidism**; give levothyroxine (case 1).
- **Man with impotence, headaches, bitemporal hemianopia, 2 cm sellar mass, PRL 3000** → **macroprolactinoma** → **cabergoline** (shrinks it; vision often recovers).
- **Man with ring no longer fitting, snoring, sweating, HTN, new DM** → **acromegaly** → IGF-1 → OGTT → MRI → surgery. Also **colonoscopy** and echo.
- **Acromegaly with a normal pituitary MRI** → look for **ectopic GHRH (bronchial carcinoid, pancreatic NET)**: CXR/CT.
- **Acromegaly + hypercalcaemia + peptic ulcers** → **MEN 1**.
- **After trans-sphenoidal surgery, clear nasal drip that is glucose/β2-transferrin positive** → **CSF leak** (meningitis risk).
- **Tall boy with small firm testes and gynaecomastia** → **Klinefelter** (karyotype).

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6. Rapid Revision (Last-Day)

1. Pituitary in the **sella**, under the **diaphragma sellae**, **½–1 g**. Anterior lobe fed by **long portal veins**; posterior by the **inferior hypophyseal artery**.
2. **Acidophils: GH, PRL. Basophils: TSH, ACTH, FSH, LH. Chromophobes: none.**
3. ADH from the **supraoptic** nucleus, oxytocin from the **paraventricular**.
4. **TRH → TSH + PRL; GnRH → LH > FSH; GHRH → GH; somatostatin ↓ GH (and insulin, glucagon, gastrin, VIP); CRH → ACTH; dopamine ↓ PRL.**
5. Tumours = **15%** of intracranial. **PRL 35 · NF 30 · GH 20 · mixed 10 · others 5**.
6. Mass effects: **bitemporal headache**, **bitemporal hemianopia**, DI/SIADH, **CN III/IV/V1/VI**, **CSF rhinorrhoea**, pyramidal signs. Acute = **apoplexy**.
7. ↑ PRL causes: **pregnancy, 3 S**; **drugs** (antipsychotics, methyldopa, reserpine, oestrogen, opiates); **3C** (CRF, cirrhosis, chest wall), **3P** (pituitary/stalk, paraneoplastic, **primary hypothyroidism**).
8. Prolactinoma: ♀ micro (amenorrhoea, galactorrhoea, osteoporosis, hirsutism); ♂ macro (↓ libido, impotence, infertility).
9. Normal PRL ♀ **< 25,** ♂ **< 20**; **< 200 → other cause**. **Hormones before imaging.**
10. Treatment: **dopamine agonist first** (bromocriptine 2.5 mg tds; **cabergoline better**; nausea, postural hypotension). Surgery: trans-sphenoidal (< 10 mm) or trans-frontal (> 10 mm). RT 4500 rads.
11. **Secretory non-PRL adenomas → surgery; asymptomatic micro / non-secretory → observe.**
12. GH: ↑ glucose, protein synthesis, FFA; via **IGF-1 (liver)**. **Hypoglycaemia, exercise, L-dopa, clonidine ↑ GH; hyperglycaemia ↓**.

13. **Gigantism before fusion; acromegaly after.** Tall stature: constitutional commonest; Klinefelter (XXY, azoospermia), Marfan, homocystinuria (MR), Sotos, eunuchoidism.
14. Acromegaly: coarse face, **prognathism**, **macroglossia**, spade hands, **↑ ring/shoe size**, **carpal tunnel**, OA, **HTN, cardiomegaly, OSA**, oily sweaty skin, **DM 25% (IGT 50%)**, hypercalcaemia 16%, co-secretion, MEN 1.
15. X-ray: double floor → ballooned sella; **tufted phalanges**; **heel pad > 22 mm**.
16. **IGF-1 = best initial; OGTT GH not < 2 ng/mL = most accurate**; MRI; CXR for carcinoid; **echo + colonoscopy**.
17. Treatment: **surgery** → octreotide (SC q12h) / DA 15–30 mg (10–40% respond) / RT (years).

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7. Exam-Style Questions

Q1 (lecture case, slides 51–52). A 32-year-old woman has irregular menses, galactorrhoea, fatigue and weight gain. She started **clozapine** 5 months ago and has **stopped levothyroxine** for 6 months. Visual fields normal. **TSH 17**, **PRL 85**. What is the most likely explanation for the nipple discharge? A. **Hypothyroidism** · B. Prolactinoma · C. Thyrotropic adenoma · D. Ectopic prolactin · E. Medication adverse effect

> **Answer: A. Hypothyroidism.** *[reasoning added; key not on slides]*
> - Untreated **primary hypothyroidism** (TSH 17, fatigue, weight gain) → ↑ TRH → ↑ prolactin.
> - PRL **85 < 200** argues against a prolactinoma, and visual fields are normal.
> - **Clozapine is the classic prolactin-SPARING antipsychotic** (weak D2 blockade), so E is the distractor.
> - A **TSH-oma** gives high T4 with inappropriately normal or high TSH, with hyperthyroid features.
> - Management: restart levothyroxine, then recheck PRL.

Q2 (lecture case, slides 53–54). A 40-year-old man has bitemporal headaches for 5 months, diplopia, impaired peripheral vision bilaterally and a pale look. MRI: **16 × 11 × 9 mm intrasellar mass**. Further evaluation is most likely to show: A. Galactorrhoea · B. Coarse facial features · C. Diffuse goitre · D. **Erectile dysfunction** · E. Abdominal striae

> **Answer: D.** *[reasoning added]*
> - A **macroadenoma in a man**: either a macroprolactinoma (the commonest functioning type, presenting late in men) or a non-functioning adenoma.
> - Both cause **hypogonadotrophic hypogonadism → ↓ libido and erectile dysfunction**.
> - **Galactorrhoea is rare in men** (A).
> - He has no acromegalic (B), thyrotoxic (C) or Cushing's (E) features. His pallor fits hypopituitarism.

Q3. Which test is the most accurate (confirmatory) for acromegaly? A. Random GH · B. IGF-1 · C. **GH after a 75 g oral glucose load** · D. Pituitary MRI · E. GHRH level

> **Answer: C.**
> - **IGF-1 is the best initial test**.
> - **Failure of GH to suppress (< 2 ng/mL) on OGTT confirms the diagnosis** (slide 47).
> - Random GH is unreliable because secretion is pulsatile.

Q4. A 28-year-old woman with amenorrhoea, galactorrhoea, PRL 260 ng/mL and a 7 mm pituitary adenoma wants to conceive. What is the first-line treatment? A. Trans-sphenoidal surgery · B. Radiotherapy · C. **Cabergoline (or bromocriptine)** · D. Octreotide · E. Observation

> **Answer: C.**
> - Dopamine agonists are first line for **symptomatic** prolactinoma (slide 27). They restore ovulation and shrink the tumour.
> - *[added]* **Bromocriptine** has the longer safety record in pregnancy.

Q5. A patient with acromegaly has a **normal pituitary MRI** and a raised **GHRH**. What should be looked for next? A. Craniopharyngioma · B. **Bronchial carcinoid (CXR/CT chest)** · C. Pheochromocytoma · D. Medullary thyroid cancer · E. Insulinoma

> **Answer: B.**
> - **Ectopic GHRH from a bronchial carcinoid** (slides 46–47).

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8. Exam Pitfalls & Traps

- **Slide 10 order error:** it says oxytocin and ADH are made in the paraventricular and supraoptic nuclei "respectively". The lecture's own posterior-pituitary slide (Lecture 11, slide 4) says the reverse: **ADH = supraoptic, oxytocin = paraventricular**. Use that.
- **Slide 12 "LH > FSH" with GnRH:** true for the pulsatile response. Don't confuse it with the LH/FSH ratio in PCOS.
- **Prolactinoma first-line is MEDICAL.** Picking surgery is the commonest error. Every other functioning adenoma → surgery.
- **PRL < 200 ng/mL** → look for drugs, hypothyroidism, renal failure or stalk compression before calling it a prolactinoma.
- **Clozapine** (and aripiprazole) are **prolactin-sparing**; **risperidone** and haloperidol raise PRL most *[added]*.
- **Acromegaly diagnosis:** IGF-1 (best initial) → **OGTT** (most accurate). A random GH > 6 ng/mL (slide 47) is **not** diagnostic alone.
- **Bromocriptine doses differ: 2.5 mg tds** for prolactinoma vs **15–30 mg/day** for acromegaly.
- **Somatostatin vs octreotide:** somatostatin is non-specific and needs an infusion; octreotide is longer acting, specific and given SC every 12 h.
- **Hypercalcaemia in acromegaly** (16%) comes from **↑ 1,25(OH)$_2$D**. Also think **MEN 1** (parathyroid adenoma).
- **Colonoscopy** is part of the acromegaly work-up (polyps, colon cancer).
- **Slide 45 typo:** "Frognathism" = **prognathism**.
- **Klinefelter:** "positive buccal smear" (Barr body), because XXY has an inactivated X.